

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**Supplementary Winter Examination – 2024****Course: Second Year B. Tech (Sem-IV) Branch : Civil Engineering Semester :IV****Subject Code & Name: BTCVC404/BTCIC404/BTCEC404 Water Resources Engineering****Max Marks: 60****Date: 27/12/2024****Duration: 3 Hr.****Instructions to the Students:**

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	What is the term used to describe the relationship between the area irrigated and the volume of water used? a) Base period b) Delta c) Duty d) Consumptive use	BL2/CO 1	1
2	Which type of irrigation system involves delivering water using gravitational flow? a) Drip irrigation b) Sprinkler irrigation c) Canal irrigation d) Lift irrigation	BL2/CO 1	1
3	What does a mass curve represent in reservoir planning? a) Rainfall data b) Cumulative inflow and outflow c) Flood frequency d) Water losses	BL2/CO 1	1
4	The useful life of a reservoir is primarily reduced by: a) Silting b) Rainfall c) Evaporation losses d) Demand curve	BL2/CO 1	1
5	Which of the following is a primary function of a gallery in a gravity dam? a) Enhancing stability b) Preventing seepage c) Drainage and inspection d) Increasing water pressure	BL2/CO 2	1
6	Which failure mode is specific to earth dams? a) Sliding b) Piping c) Overturning d) Arching	BL2/CO 1	1
7	The Bligh's creep theory is used to analyze: a) Silt deposition b) Seepage in weirs c) Stability of dams d) Alignment of canals	BL2/CO 2	1
8	What is the primary function of inverted filters in earth dams? a) To increase drainage efficiency b) To reduce uplift pressure c) To minimize seepage d) To support structural stability	BL2/CO 1	1
9	The unit hydrograph is used to determine: a) Rainfall intensity b) Catchment area size c) Runoff response from rainfall d) Base flow discharge	BL2/CO 2	1
10	Which formula is typically used to estimate runoff in a catchment?	BL2/CO 2	1

	a) Darcy's equation	b) Manning's formula	c) Rational method	d) Bernoulli's equation		
11	Flood hydrographs are primarily used to study:				BL2/CO 3	1
	a) Evaporation losses	b) Rainfall distribution	c) Flow variations during a flood	d) Groundwater recharge		
12	Which type of reservoir classification is based on its purpose?				BL2/CO3	1
	a) Single-purpose	b) Zoned reservoir reservoir	c) Overflow reservoir	d) Silting reservoir		
Q. 2	Solve the following.					12
A)	Derive relationship between duty and delta.				BL2/CO 1	6
B)	Differentiate between the lift irrigation and flow irrigation				BL2/CO1	6
Q.3	Solve the following.					12
A)	Explain the process of selecting a site for a reservoir. What are the key factors that influence the site selection for constructing a reservoir?				BL3/CO2	6
B)	Explain elementary profile of gravity dam.				BL3/CO1	6
Q. 4	Solve Any Two of the following.					12
A)	Discuss the components of an earthen dam, their functions, and the design criteria to ensure stability and prevent failure.				BL3/CO2	6
B)	Explain Bligh's creep theory and state its limitations.				BL2/CO2	6
C)	Determine the section of Lacey's regime channel to carry 15 m ³ /s of water in an alluvium with a sediment size of 0.75 mm.				BL4/CO3	6
Q.5	Solve Any Two of the following.					12
A)	Explain the hydrologic cycle and describe the major processes involved in it.				BL2/CO2	6
B)	What is a flood hydrograph, and how is it used to estimate flood flow in a river or stream?				BL2/CO2	6
C)	In a catchment area, there are five existing rain gauge stations. The annual rainfall at these stations is recorded as follows: 800 mm, 720 mm, 600 mm, 850 mm, and 780 mm. The allowable percentage error in the mean rainfall value is 12%. Calculate the optimum number of rain gauges required in the catchment to achieve the desired accuracy.				BL2/CO1	6
Q. 6	Solve Any Two of the following.					12
A)	Describe the causes of water logging in irrigated lands. Discuss the preventive and curative measures for it.				BL2/CO3	6
B)	Explain with neat sketch rain water harvesting.				BL3/CO3	6
C)	A tube well is constructed in a basin with a specific yield of 0.25. If the area of influence of the well is 1000 m ² and the drawdown during pumping is 10 m, calculate the discharge of the tube well. Use Darcy's Law for the calculation.				BL4/CO3	6
*** End ***						